

ECE 6258 Digital Image Processing Project Report

Topic: Digital Watermarking

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There are several techniques in digital watermarking, such as spatial domain encoding, frequency domain embedding, and wavelet domain embedding, etc. My project was implemented in DCT domain, which has several advantages such as robustness in JPEG compression, and identity for the original image and the watermarked image. I also did a brief test in the wavelet domain and compared it with my results obtained in DCT domain. The coding work was done in MATLAB.

Embedding Algorithm

First, the images in the project database are converted from RGB to gray-scale to be the original host images, and the watermark is converted into binary image using a gray level threshold. Now the images are ready to go.

The block-based DCT is performed with the host image by 8x8 blocks of data, and the variance of each block is calculated. For those blocks whose variance are above a certain threshold, a series of watermark values are embed in 9 positions (marked with black dot in Fig.1) in the 8x8 block. The selection of these positions is a tradeoff between impact on the host image and robustness in image compression. And the reason for giving a threshold of variance is because if the watermark were embedded in an area with small variance (low frequency area), it would cause the visible difference between original host image and the watermarked image; and if the watermark were embedded in an area with high variance (high frequency area), the watermarked image will be visually identical to the original image even there are actually slight differences between them.

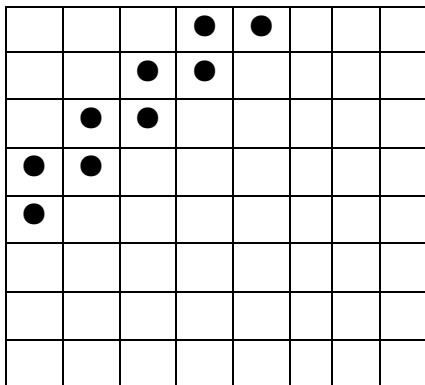


Fig.1 positions to embed watermark in an 8x8 block (DCT domain)

The embedding algorithm could be described as $X = S(1 + \alpha W)$, where S is the DCT transform of original host image, X is the watermarked signal, and W is the watermark, which is the Gatech logo in this project. α is a scaling vector.

The result of this embedding procedure is a new DCT domain signal consists of 8x8 blocks that have been watermarked or left unmodified from the DCT of original host image. The positions of embedded blocks are recorded by a flag matrix and saved to disk.

This DCT domain 2-D signal is then performed 8x8 block-based inverse DCT to obtain the watermarked image.

The watermarked image is saved to disk as bitmap format (.bmp file), JPEG format (.jpg file) at 2bpp, 1bpp, 0.5bpp, and 0.25bpp respectively for later extracting test.

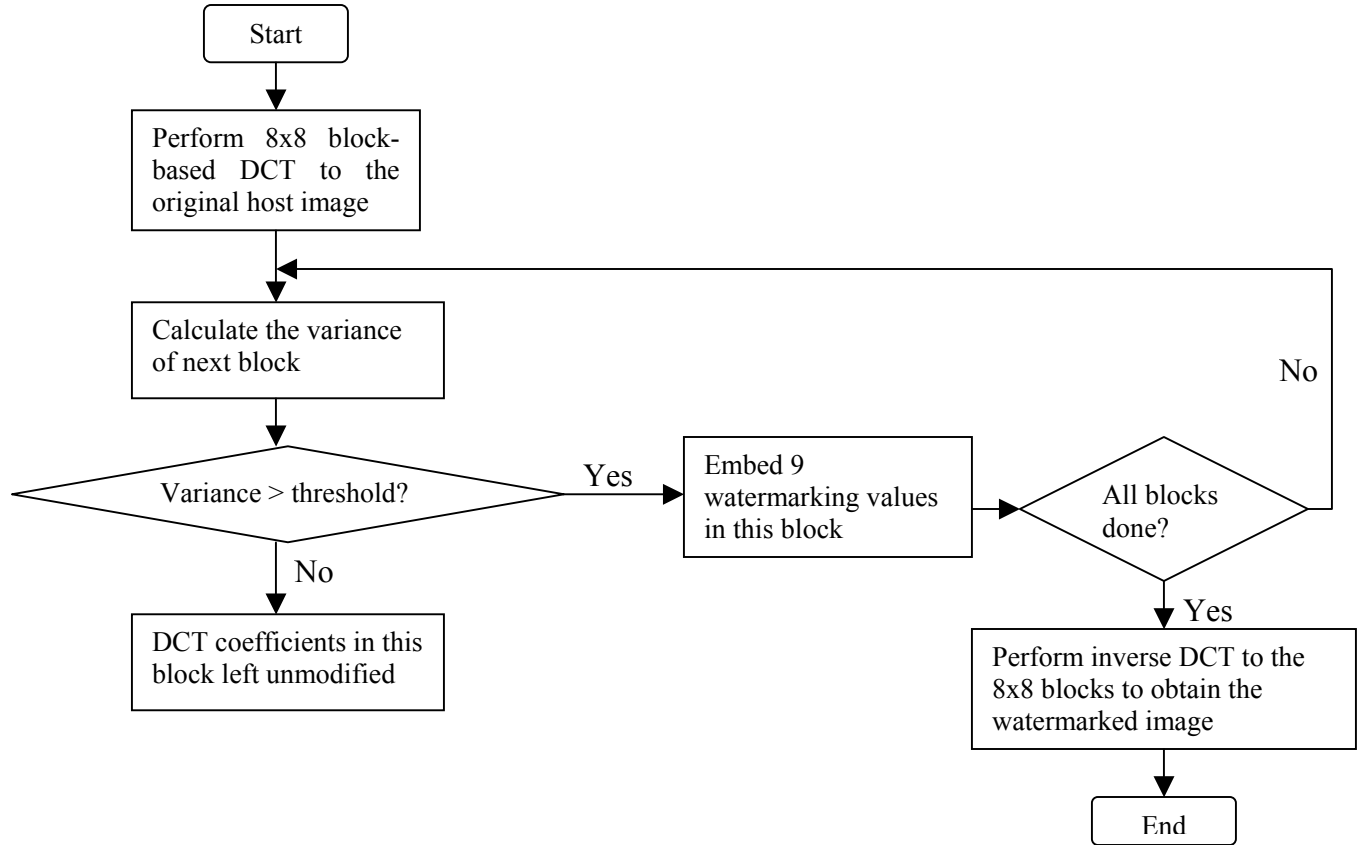


Fig.2. Flow chart of the watermark embedding procedure

Watermark Extraction Procedure

Now it's time to see if the watermark is still alive in the watermarked and compressed images. First read in the original host image and the watermarked image, and perform DCT transform on both of them. Subtract the original host image from the watermarked image by using $W = \frac{X}{S} - 1$, where 1 is a matrix full of ones. Now load the flag that has the record of which blocks were embedded with watermark, and find out which parts of W belong to the watermark before. These elements are first taken to be connected and form a watermark vector, and then are rearranged to a 2D image of recovered watermark image. This image is finally divided by a scaling factor to give better contrast.

Results

The above watermarking algorithm was tested with 3 images in project database, which are honolulu.jpg, Kalalau1185.jpg, and haleakala.jpg respectively. The binary watermark is the Georgia tech logo which is shown in Fig.3.(a), the extracted logo from compressed images are shown in Fig.3, Fig.4, Fig.5 for different host images.

1. honolulu

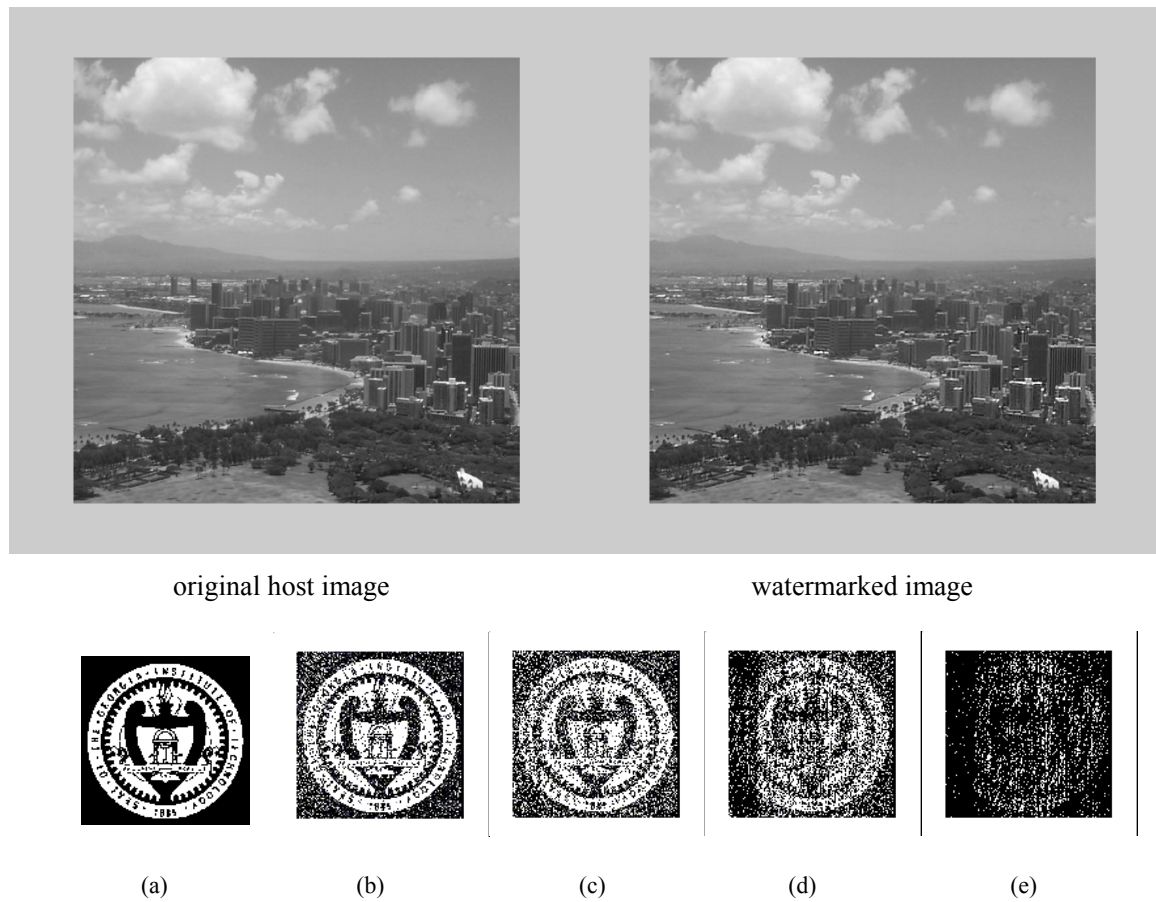
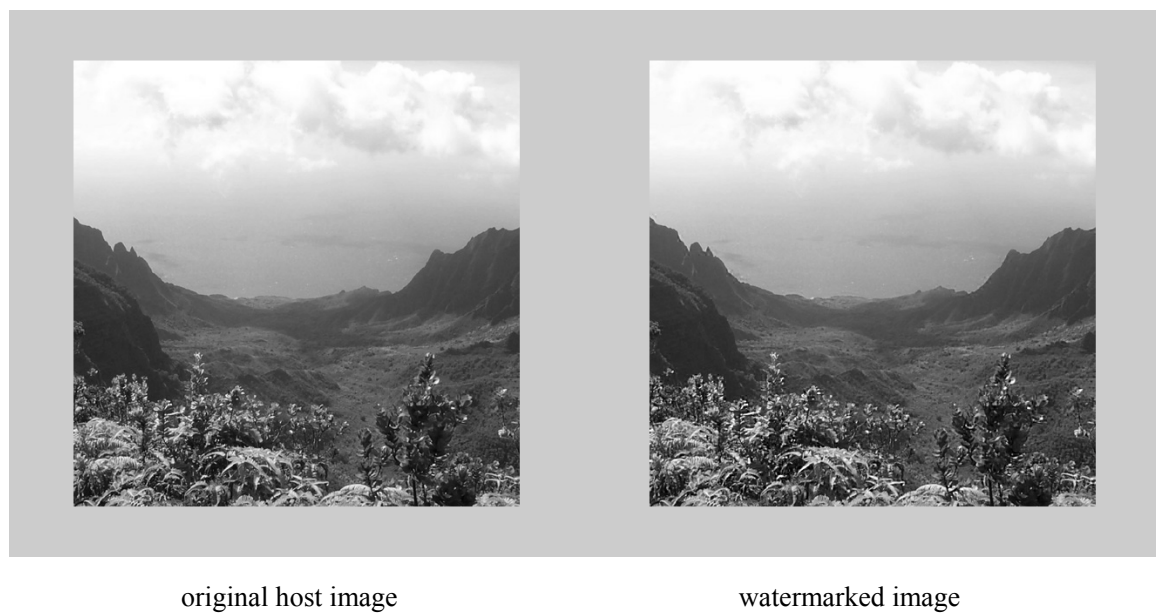


Fig.3. Testing results for honolulu.jpg. (a) original binary logo; (b) logo extracted from compressed watermarked image of 2bits/pixel; (c) 1 bit/pixel; (d) 0.5 bits/pixel; (e) 0.25 bits/pixel.

2. Kalalau1185



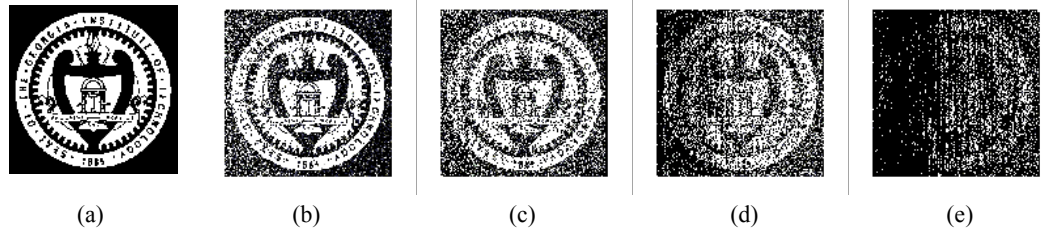


Fig.4. Testing results for Kalalau1185.jpg. (a) original binary logo; (b) logo extracted from compressed watermarked image of 2bits/pixel; (c) 1 bit/pixel; (d) 0.5 bits/pixel; (e) 0.25 bits/pixel.

3. haleakala

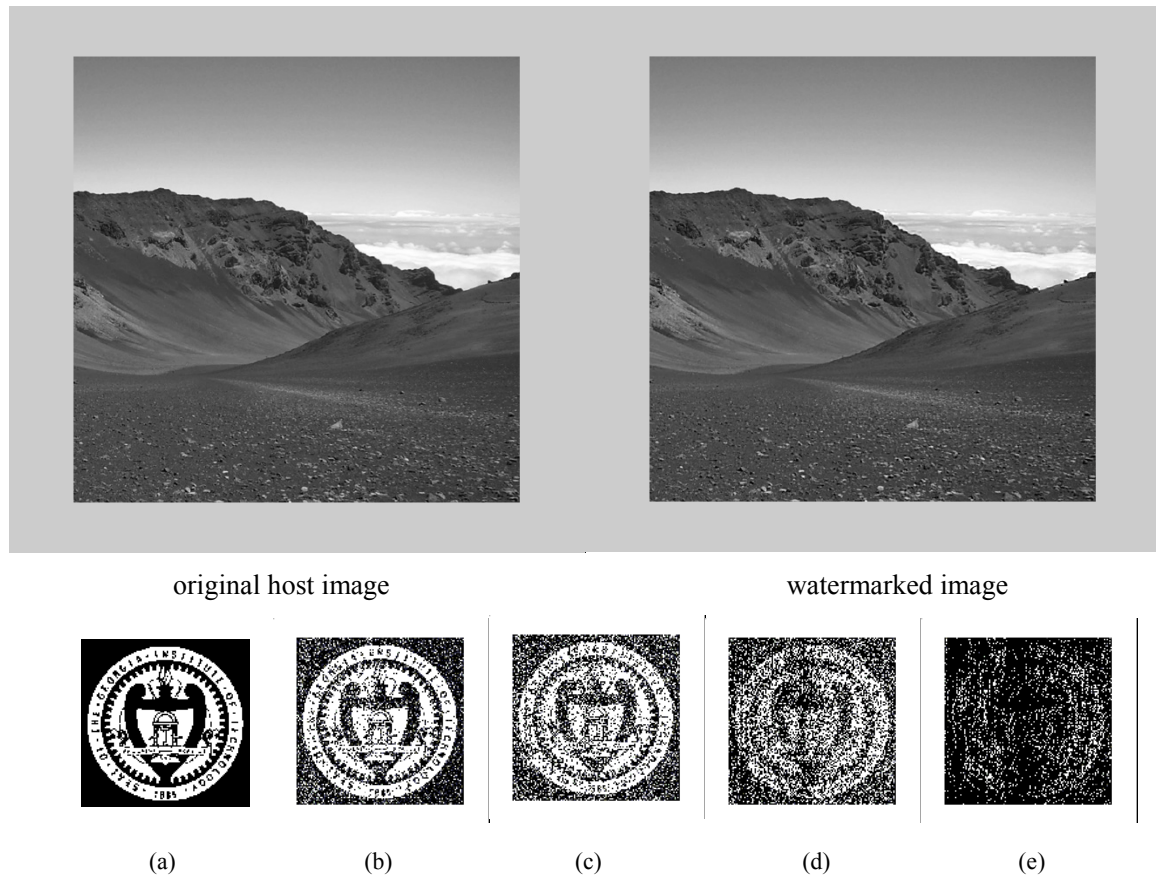


Fig.5. Testing results for haleakala.jpg. (a) original binary logo; (b) logo extracted from compressed watermarked image of 2bits/pixel; (c) 1 bit/pixel; (d) 0.5 bits/pixel; (e) 0.25 bits/pixel.

Discussion and further testing

The block-based DCT domain embedding algorithm did well in digital watermarking, and it can meet the requirements of 1) the watermarked images are visually identical to the original images; 2) the watermark can be extracted from compressed images with standard JPEG algorithm, especially when the compress rate is above 0.5 bits/pixel. This algorithm was tested by different images, which showed satisfying results. No specific knowledge about the particular image or watermark was used.

It is also a popular technique to implement digital watermarking in wavelet domain, where the watermark is most likely to be embedded in HL and LH sub bands to keep watermark robust and reduce visual degradation.

Fig.6 shows the testing result on image Kalalau1185.jpg, using wavelet domain embedding algorithm. This test is a rather simple one by doing DWT with 'haar' filter and directly adding the scaled watermark to HH (high-high frequency) sub band, which may cause visible difference between original and watermarked image. But it still shows acceptable result in identity between the original and watermarked image, and robustness with JPEG when compressed rate is above 0.5 bits/pixel. More complicated and efficient algorithm in this domain is needed to get better results.

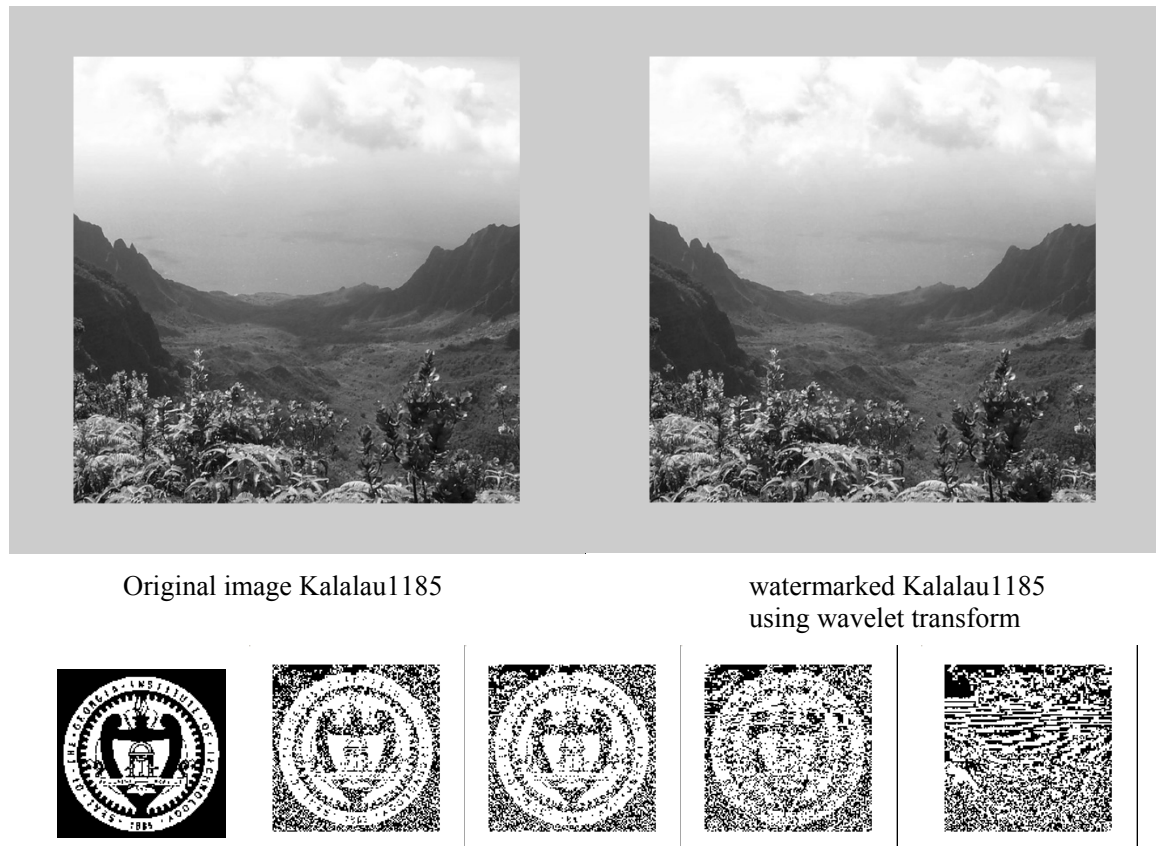


Fig.6. Testing results for Kalalau1185.jpg using wavelet transform. (a) original binary logo; (b) logo extracted from compressed watermarked image of 2bits/pixel; (c) 1 bit/pixel; (d) 0.5 bits/pixel; (e) 0.25 bits/pixel.

Both DCT domain and wavelet domain embedding are efficient techniques in digital watermarking. Embedding in DCT domain is a mature technique that has been proved to be a good candidate for watermarking, for both reasons of robustness as well as visual impact, and embedding in wavelet domain is a relatively new one, but shows a lot of potentials in digital watermarking.